MIL-A-26442(USAF)
20 January 1959

MILITARY SPECIFICATION

ADAPTERS, WAVEGUIDE-TO-COAXIAL

1. SCOPE

1.1 SCOPE.- This specification covers three classes of waveguide-to-coaxial adapters hereinafter referred to as the "adapters".

1.2 CLASSIFICATION.- The adapters shall be of the following classes and types, as specified (see 6.2):

Class I - Waveguide to Type N

Type 1 - UG-397( )/U
Type 2 - UG-398( )/U
Type 3 - UG-399( )/U
Type 4 - UG-400( )/U
Type 5 - UG-446( )/U
Type 6 - UG-952( )/U
Type 7 - UG-953( )/U

Class II - Waveguide to 5/8-inch Coaxial Line

Type 1 - UG-403( )/U
Type 2 - UG-404( )/U
Type 3 - UG-445( )/U
Type 4 - UG-520( )/U
Type 5 - UG-521( )/U
Type 6 - UG-523( )/U

Class III - Waveguide to 7/8-inch Coaxial Line

Type 1 - UG-405( )/U
Type 2 - UG-522( )/U

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-W-85	Waveguides, Rigid, Rectangular
MIL-F-3922	Flanges, Waveguide, And Associated Fittings

FSC 5935

THE REPUBLICAN PRESS, Hamilton, Ohio
May 1, 1959 500

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MIL-D-5028	Drawings and Data Lists; Preparation Of Manufacturer's (For Production Aeronautical And Associated Equipment)
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation For Delivery
MIL-T-9107	Test Reports, Preparation Of
MIL-F-14072	Finishes For Ground Signal Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MS15507	Connector, Adapter (Radio Frequency Adapter UG-57B/U) Series N

DRAWINGS

Armed Services Electro Standards Agency

AS-2022	Adapter Connector UG-397/U Adapts RG-51/U Waveguide To Type "N" Plug
AS-2023	Adapter Connector UG-398/U Adapts RG-50/U Waveguide To Type "N" Plug
AS-2024	Adapter Connector UG-521/U Adapts RG-49/U Waveguide To 5/8-Inch Coaxial Line
AS-2026	Adapter Connector UG-520/U Adapts RG-50/U Waveguide To 5/8-Inch Coaxial Line
AS-2027	Adapter Connector UG-445/U Adapts RG-48/U Waveguide To 5/8-Inch Coaxial Line
AS-2028	Adapter Connector UG-403/U Adapts RG-50/U Waveguide To 5/8-Inch Coaxial Line
AS-2029	Adapter Connector UG-523/U Adapts RG-48/U Waveguide To 5/8-Inch Coaxial Line
AS-2030	Adapter Connector UG-405/U Adapts RG-48/U Waveguide To 7/8-Inch Coaxial Line

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AS-2032	Adapter Connector UG-399/U Adapts RG-49/U Waveguide To Type "N" Plug
AS-2033	Adapter Connector UG-404/U Adapts RG-49/U Waveguide To 5/8-Inch Coaxial Line
AS-2035	Adapter Connector UG-522/U Adapts RG-48/U Waveguide To 7/8-Inch Coaxial Line
AS-2038	Adapter Connector UG-400/U Adapts RG-48/U Waveguide To Type "N" Plug

Air Force

54B54901	Adapter Connector UG-446A/U - Assembly
55C54274	Adapter, Coaxial To Waveguide UG-592/U
55D54202	Adapter, Coaxial To Waveguide UG-593/U

Navy

RE49F210	Coupling For 7/8-Inch Stub Line Consisting of Type UG-45/U Male Assembly and UG-46/U Female Assembly
RE49F271	Coupling For 5/8-Inch Stub Line Consisting of Type UG-140/U Male Assembly and UG-141/U Female Assembly

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to ANA Bulletin No. 143.

3.3 INTERCHANGEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-5028.

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3.4 MARKING AND IDENTIFICATION.- The adapters shall be marked in accordance with MIL-STD-130. The following special markings shall be added:

a. Frequency range
b. Average voltage standing wave ratio (VSWR)

3.5 DESIGN AND CONSTRUCTION.- Each adapter shall consist of a coaxial connector mounted perpendicularly on the broad face of a short waveguide section. The center conductor of the coaxial connector shall extend into the waveguide as a probe. A waveguide flange shall be provided at the input end of the waveguide. The adapters shall be in accordance with the drawings specified in table I. When the requirements of this specification and the drawings conflict, the requirements of this specification shall govern.

3.5.1 WAVEGUIDES.- Waveguides shall be of the types as specified in table I and shall be in accordance with MIL-W-85.

3.5.2 WAVEGUIDE FLANGES.- Waveguide flanges shall be of the types as specified in table I and shall be in accordance with MIL-F-3922.

3.5.3 COAXIAL CONNECTORS.- Coaxial connectors shall be of the types so as to mate with the connectors specified in table I.

3.6 PERFORMANCE.- The adapters shall connect and adapt between waveguides and coaxial lines. The adapters shall provide a low VSWR connection and negligible insertion loss. The frequency range and maximum VSWR for each adapter shall be as specified in table I. The nominal coaxial impedance shall be 50 ohms.

3.7 FINISHES.- The flange faces, inside surfaces of waveguides, and all other electrical contacting surfaces shall be silver plated and rhodium plated over the silver to a thickness of 0.000010 plus or minus 0.000005 inch. External surfaces shall be finished in accordance with MIL-F-14072 except that wrinkle finishes shall not be used. The final paint coat shall be gloss gray, color No. 16376 in accordance with FED-STD-595.

3.8 GOVERNMENT-LOANED PROPERTY.- When the contract or purchase order so provides, the Government will loan to the contractor upon his request, one of each type of adapter the contractor is required to furnish (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the adapters shall be classified as follows:

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TABLE I

INDIVIDUAL ADAPTER REQUIREMENTS

Adapter Type	Frequency Range (megacycles) (mc)	Maximum VSWR	Adapter Drawing	Waveguide Type	Flange Type	Connector Shall Mate With
Waveguide to Type N						
UG-953( )/U	1120 - 1700	1.20	55D54202	RG-69( )/U	UG-417( )/U	UG-57( )/U
UG-952( )/U	1700 - 2600	1.30	55C54274	RG-104( )/U	UG-435( )/U	UG-57( )/U
UG-400( )/U	2600 - 3950	1.28	AS-2038	RG-48( )/U	UG-53( )/U	UG-57( )/U
UG-399( )/U	3950 - 5850	1.32	AS-2032	RG-49( )/U	UG-149( )/U	UG-57( )/U
UG-398( )/U	5850 - 8200	1.32	AS-2023	RG-50( )/U	UG-344( )/U	UG-57( )/U
UG-397( )/U	7050 - 10000	1.35	AS-2022	RG-51( )/U	UG-51( )/U	UG-57( )/U
UG-446( )/U	8200 - 12400	1.30	54B54901	RG-52( )/U	UG-39( )/U	UG-57( )/U
Waveguide to 5/8-inch Coaxial Line						
UG-445( )/U	2600 - 3950	1.25	AS-2027	RG-48( )/U	UG-53( )/U	UG-141( )/U
UG-523( )/U	2600 - 3950	1.25	AS-2029	RG-48( )/U	UG-53( )/U	UG-140( )/U
UG-404( )/U	3950 - 5850	1.25	AS-2033	RG-49( )/U	UG-149( )/U	UG-140( )/U
UG-521( )/U	3950 - 5850	1.25	AS-2024	RG-49( )/U	UG-149( )/U	UG-141( )/U
UG-403( )/U	5850 - 8200	1.25	AS-2028	RG-50( )/U	UG-344( )/U	UG-140( )/U
UG-520( )/U	5850 - 8200	1.25	AS-2026	RG-50( )/U	UG-344( )/U	UG-141( )/U
Waveguide to 7/8-inch Coaxial Line						
UG-405( )/U	2600 - 3950	1.25	AS-2030	RG-48( )/U	UG-53( )/U	UG-45( )/U
UG-522( )/U	2600 - 3950	1.25	AS-2035	RG-48( )/U	UG-53( )/U	UG-46( )/U

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a. Acceptance tests . . . . . See 4.2
b. Preproduction tests . . . . . See 4.5

4.2 ACCEPTANCE TESTS.- Acceptance tests shall consist of individual tests only.

4.2.1 INDIVIDUAL TESTS.- Each adapter shall be subjected to the following tests as described under 4.4 "TEST METHODS" of this specification:

a. Examination of product
b. Functional tests

4.3 TEST CONDITIONS.- Unless otherwise specified herein, all tests shall be conducted at prevailing ambient conditions.

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT.- Each adapter shall be given a thorough mechanical and visual inspection and test to determine that the quality of material and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness
b. Nameplates, identification markings, and labels
c. Finishes
d. Joints
e. Dimensions
f. Center contacts
g. Other visual defects

4.4.1.1 GAGE TESTS FOR TYPE N CONNECTORS.- Each type N connector (class I adapters) shall be subjected to the following gage tests:

a. After insertion and removal of a 0.084-inch minimum diameter pin to a minimum depth of 3/16 inch; all four adapter contact members shall make contact within 1/32 inch of their tip ends with a 0.058-inch maximum diameter pin inserted to a minimum depth of 3/16 inch.

b. All adapter contact members shall make contact within 1/32 inch of their tip ends with a 0.065-inch minimum diameter pin inserted to a minimum depth of 3/16 inch.

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c. With a 0.065-inch minimum diameter pin inserted 3/16 inch, the contact shall pass through a 0.122-inch maximum diameter hole of 1/2 inch minimum length when a maximum pressure of 2 pounds is applied.

d. The connector shall properly mate with Straight Adapter UG-57( )/U (MS15507).

4.4.1.2 GAGE TESTS FOR 5/8-INCH CONNECTORS.- Each 5/8-inch connector (class II adapters) shall be subjected to the following gage tests:

a. Male 5/8-inch connectors shall properly mate with a type UG-141( )/U female connector (RE49F271).

b. Female 5/8-inch connectors shall properly mate with a type UG-140( )/U male connector (RE49F271).

4.4.1.3 GAGE TESTS FOR 7/8-INCH CONNECTORS.- Each 7/8-inch connector (class III adapters) shall be subjected to the following gage tests:

a. Male 7/8-inch connectors shall properly mate with a type UG-46( )/U female connector (RE49F210).

b. Female 7/8-inch connectors shall properly mate with a type UG-45( )/U male connector (RE49F210).

4.4.1.4 MECHANICAL INSPECTION TESTS FOR WAVEGUIDES AND WAVEGUIDE FLANGES.- Waveguides and waveguide flanges shall be visually and mechanically examined in accordance with the applicable test requirements of MIL-W-85 and MIL-F-3922 respectively.

4.4.2 FUNCTIONAL TESTS.- Each adapter shall be subjected to a thorough electrical test to determine compliance with the requirements of this specification.

4.4.2.1 TEST EQUIPMENT.- The following test equipment, or equal, shall be used to test the adapter:

Item	Equipment	Characteristics
1	Radio-frequency (r-f) signal generator	Capable of generating a 1000 cycle modulated r-f signal over test frequency range
2	Attenuator	Attenuation of 10 decibels (db) minimum, VSWR of 1.15 maximum
3	Waveguide and coaxial slotted lines	VSWR of 1.03 maximum, slope of 0.01 maximum with VSWR of 1.03

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Item	Equipment	
4	Tuning stub	Capable of providing a matched load, VSWR of 1.02 maximum
5	Waveguide and coaxial terminations	VSWR of 1.10 maximum for waveguide termination and 1.15 maximum for coaxial termination
6	Standing wave amplifier	High gain, low noise, capable of receiving a 1000 cycle signal
7	R-f probes with crystal detectors	Capable of detecting r-f signal over test frequency range
8	Wavemeter	Accurate to within 0.1 percent over test frequency range

4.4.2.2 TEST PROCEDURE.- Connect the test equipment and adapter to be tested as shown in figure 1 and perform the following test:

a. Apply a 1000 cycle modulated r-f signal from the signal generator (1) to the test set-up. Tune the tuning stub (4) to produce a matched load at the termination (5); position the input r-f probe (7i) to minimum depth in the slotted line (3i); and tune for maximum sensitivity.

b. Connect output probe (7o) to standing wave amplifier; move probe along slotted line (3o) to position which gives maximum meter reading on standing wave amplifier; tune standing wave amplifier for full scale deflection of meter; move probe to position which gives minimum meter reading; and read VSWR on the meter. The VSWR shall not exceed 1.02 for this matched condition.

c. Reconnect the input probe to the standing wave amplifier and repeat step "b". The VSWR shall not exceed the applicable value specified in table I.

d. Conduct this test at each of five separate frequencies evenly spaced over the applicable frequency range (see table I) using both the waveguide and connector ends of the adapter as the input end (total of 10 measurements per adapter). The following conditions shall be observed during the test:

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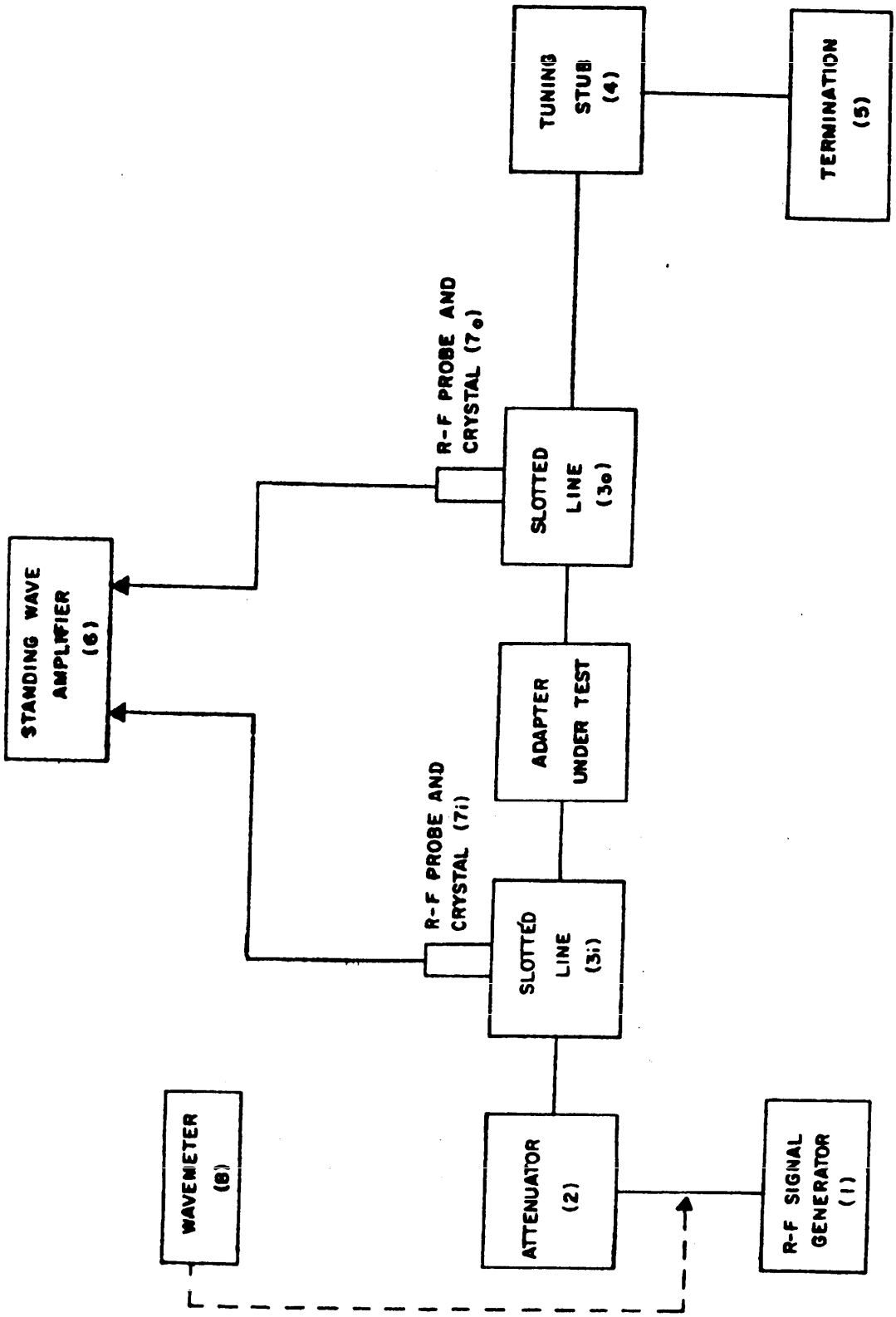


FIGURE 1. TEST SET-UP FOR ADAPTER VSWR MEASUREMENTS

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(1) The standing wave amplifier shall be run on almost full gain.

(2) The probe depth in the slotted line shall be small and kept constant.

(3) The tuning stub shall be adjusted for maximum power transfer.

4.5 PREPRODUCTION TESTS

4.5.1 PREPRODUCTION TEST SAMPLES.- When so stated in the contract, the preproduction test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or at the contractor's plant under supervision of the procuring activity.

4.5.2 PREPRODUCTION TEST REPORTS.- When the contractor performs the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.5.3 PREPRODUCTION TESTS.- Preproduction tests shall consist of all the tests described under 4.4 "TEST METHODS".

5. PREPARATION FOR DELIVERY

5.1 PREPARATION FOR SHIPMENT.- The adapters shall be prepared and marked for shipment in accordance with MIL-P-7936.

6. NOTES

6.1 INTENDED USE.- The adapters are intended for use in providing a transition between waveguides and coaxial lines.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Classes and types of adapters required (see 1.2).

c. Conditions under which Government-loaned property will be made available to the contractor (see 3.8).

d. Number of preproduction samples required (see 4.5.1).

e. Location and inspection of preproduction tests (see 4.5.1).

f. Applicable levels of preservation, packaging, and packing required (see 5.1).

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6.3 NOMENCLATURE.- The parenthesis appearing in the nomenclature will be dropped or replaced by a letter to designate the particular design. Nomenclature for the individual components, where required, will be furnished upon application to the procuring activity.

6.4 DRAWINGS.- Upon specific request of the contractor to the procuring activity, a copy of the applicable drawings, if available, may be furnished for informational purposes only.

NOTICE.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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